ADP-Based Fault-Tolerant Control for Nonlinear Systems

Project Summary | Methodology, Numerical Results, and Implementation Scope

Abstract

This project combines an adaptive observer with offline approximate dynamic programming (ADP) to regulate a continuous-time nonlinear system subject to additive actuator and sensor faults. A virtual output removes the modeled sensor-fault channel from the observer input; actuator-fault estimates support control compensation, while the sensor fault is reconstructed algebraically. A single polynomial critic is fitted by least-squares policy iteration using the known dynamics. The stored training run reaches its stopping criterion after four iterations. In the matched amplitude-1 experiment, compensation reduces state integral-square error by approximately 45.0% and finite-horizon cost by 28.8% relative to the uncompensated controller. Sensor-only reconstruction achieves an active-interval RMSE of 1.97×10^{-5}. These findings describe the supplied deterministic simulation example and its stated assumptions.

1. Problem and demonstration model

The objective is regulation about the origin while estimating states and faults. Actuator faults enter the state equation, and sensor faults corrupt measured outputs. The mathematical model is

$$\dot{x} = Ax + Bu + g(x) + D_a f_a, \qquad y = Cx + D_s f_s$$

Table 1. Numerical plant and cost definition.

Quantity	Value
State matrix A	[0, 1; -2, -0.8]
Input and actuator-fault channels	$B = D_a = [0; 1]$
Measurement matrix C	[1, 0; 0, 1; 1, 1]
Sensor-fault channel D_s	[0; 0; 1]
Known nonlinear term g(x)	$[0; 0.25 \sin(x_1)]$
Cost weights	$Q_c = \mathrm{diag}(4, 1)$; R = 0.5
Design domain	$-2 \le x_1 \le 2$; $-2 \le x_2 \le 2$

The example has two states, one control input, three outputs, and one fault channel of each type. The actuator fault is matched to the control input. Only the third output is sensor-fault affected. The nonlinearity has Lipschitz constant 0.25.

Evidence basis. *Principal results use the parameter-matched archive results/1/. Color figures are reproduced directly from the Persian report, retaining their original plots, labels, and colors. The separate amplitude-sweep archive is identified explicitly. No new MATLAB simulation is claimed.*

2. Observer, fault reconstruction, and control

The sensor-fault projection is N_s = [1, 0, 0; 0, 1, 0]. It satisfies $N_s D_s = 0$, giving $C_0 = N_s C = I_2$. Thus y_0 = x in this example: both states are directly available through healthy output channels. This favorable arrangement is central to interpreting the numerical evidence.

$$r = y_0 - C_0\hat{x}, \qquad y_0 = N_s y = C_0 x$$

$$\dot{\hat{x}} = A\hat{x} + Bu + g(\hat{x}) + D_a\hat{f}_a + Lr$$

$$\dot{\hat{f}}_a = -\delta\Gamma\hat{f}_a + \Gamma K r$$

The observer uses adaptation gain Γ = 3 and leakage δ = 0.25. The gains satisfy a Lipschitz observer LMI with P positive definite, $L = P^{-1}Y$, and the equality $KC_0 = D_a^T P$. The implementation solves these constraints using YALMIP when available and numerically checks the result.

$$u = \mu_N(\hat{x}) - B^{\dagger} D_a \hat{f}_a, \qquad \hat{f}_s = D_s^{\dagger}(y - C\hat{x})$$

The final accepted policy μ_N is evaluated at the estimated state. For $B = D_a$, the compensation term simplifies to $-f_a$. Sensor reconstruction supports monitoring; the projected output provides sensor-fault decoupling in the feedback path. Its estimation error obeys $\tilde{f}_s = -D_s^{\dagger} C e_x$, where $e_x = \hat{x} - x$.

Table 2. Structural checks for the supplied plant.

Condition	Verified value
Full column rank of D_s	$\mathrm{rank}(D_s) = 1 = n_s$
Output redundancy	$n_y = 3 > n_s = 1$
Sensor projection / actuator matching	$\lVert N_s D_s \rVert = 0$; $\lVert D_a - BB^{\dagger} D_a \rVert = 0$
Adaptive-gain necessary rank condition	$\mathrm{rank}(C_0 D_a) = 1$
Sampled Rosenbrock matrix	Minimum rank 3; minimum singular value 0.067265

The sampled complex grid is diagnostic. Here $C_0 = I_2$ and D_a has full column rank, which also establishes the Rosenbrock full-rank property for every complex s.

Table 3. Stored observer feasibility diagnostics.

Diagnostic	Value
Largest eigenvalue of LMI matrix	-1.75317
Smallest eigenvalue of P	1.36833
Adaptive equality residual, Frobenius norm	5.97×10^{-10}
LMI scalar ρ	3.03778

The archive records a solver-based observer, with L approximately [1.24045, -0.48745; -0.49368, 0.45105]. An analytical fallback is available for the supplied $C_0 = I_2$ arrangement, but it produces different gains and may change the trajectories.

3. Offline ADP policy iteration

Training uses a single eight-feature polynomial critic and derives the nominal policy from its gradient and the known input matrix.

$$\hat{V}(x) = W^T \sigma(x), \qquad \mu_{\text{cand}}(x) = -\tfrac{1}{2} R^{-1} B^T J_\sigma(x)^T W$$

$$\sigma = [z_1^2,\ z_1 z_2,\ z_2^2,\ z_1^4,\ z_1^3 z_2,\ z_1^2 z_2^2,\ z_1 z_2^3,\ z_2^4]^T, \quad z_i = x_i/2$$

The 17 × 17 grid excludes the origin, giving 288 collocation points. Starting from $\mu_0(x) = -[1.6,\ 1.2]x$, each iteration rebuilds and solves the frozen-policy Bellman regression:

$$\Omega_{k,:} = [J_\sigma(x_k)(Ax_k + g(x_k) + B\mu_i(x_k))]^T$$

$$b_k = x_k^T Q_c x_k + \mu_i(x_k)^T R \mu_i(x_k), \qquad W_i = \arg\min_W \|\Omega W + b\|_2^2$$

Rank must equal eight. Candidates are screened with a quadratic Lyapunov function from the initial policy. Blend coefficients 1, 1/2, ..., 1/64 are tested against a sampled decay margin of 10^{-3}; failed candidates leave the previous policy unchanged.

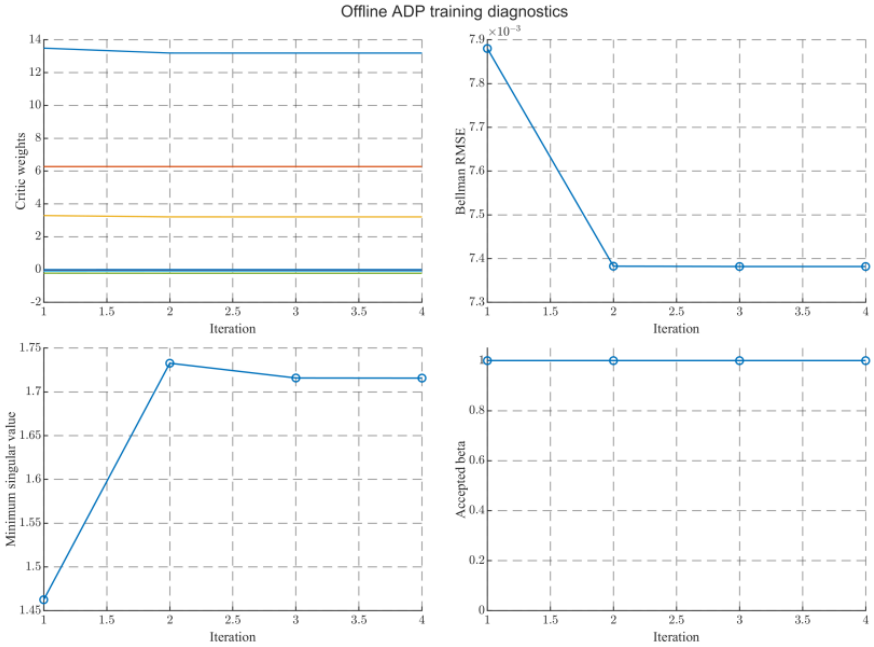


Figure 1. Original ADP diagnostics: critic weights, Bellman RMSE, minimum singular value, and accepted blend coefficient. Reproduced from the Persian report, Figure 4-1.

Table 4. Numerical training history.

Iteration	Bellman RMSE	Rank	Min. singular value	Condition no.	β
1	0.007880	8	1.4625	77.42	1.0
2	0.007383	8	1.7328	70.32	1.0
3	0.007382	8	1.7159	70.46	1.0
4	0.007382	8	1.7158	70.46	1.0

All candidates passed without blending; the final sampled decay margin was 0.92241. The weight-change tolerance is 10^{-5}, with 3-20 iterations allowed. The final policy is fixed during simulation. Numerical convergence and grid screening do not establish uniform admissibility or a suboptimality bound.

4. Experimental setup and evaluation measures

Table 5. Common simulation settings.

Setting	Value
Initial plant / observer states	$x(0) = [1.2; -0.7]$; $\hat{x}(0) = [0; 0]$
Initial actuator-fault estimate	$f_a(0) = 0$
Time horizon / recorded interval	0-25 s / 0.01 s (2,501 output samples)
Solver / maximum step	ode45 / 0.02 s
Relative / absolute tolerance	10^{-7} / 10^{-9}
Measurement noise / input limit	Noise-free / $u_{max} = \infty$

Table 6. Four proposed-controller scenarios.

ID	Actuator fault	Sensor fault
1	Zero	Zero
2	Smooth actuator profile	Zero
3	Zero	Piecewise-constant bias
4	Smooth actuator profile	Piecewise-constant bias

The actuator profile is $f_a(t) = [1 + 0.10 \sin(0.8(t - 5))][s(t - 5) - s(t - 17)]$, where $s(\tau) = [1 + \tanh(3\tau)]/2$. Transitions are centered at 5 and 17 s. Sensor bias is 0 before 8 s, 0.45 until 16 s, and 0.20 thereafter.

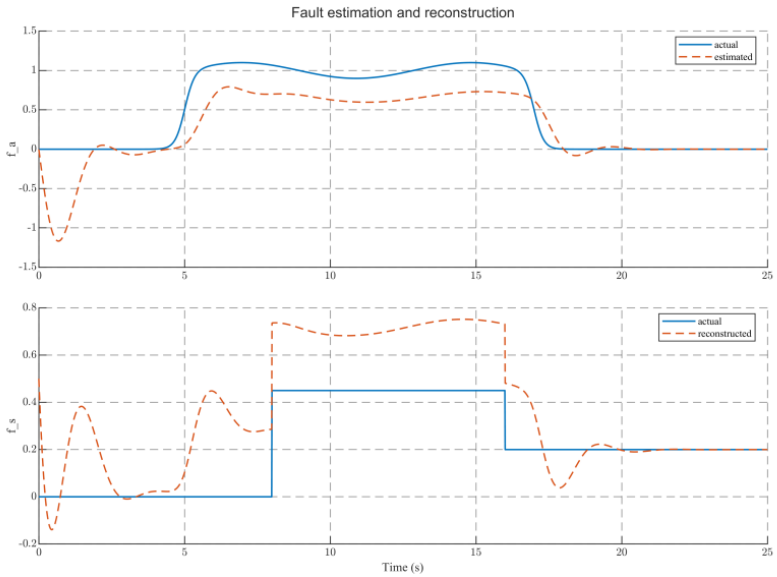


Figure 2. Actual and estimated actuator and sensor faults in the simultaneous-fault scenario. Reproduced from the Persian report, Figure 4-16. The sensor bias remains active after 16 s.

$$\mathrm{ISE}_x = \int_0^T \|x(t)\|_2^2 dt, \quad E_u = \int_0^T \|u(t)\|_2^2 dt, \quad J_T = \int_0^T (x^T Q_c x + u^T R u) dt$$

State-estimation RMSE is the square root of the sample mean of $\|\hat{x} - x\|^2$. Fault RMSE is evaluated where the true fault norm exceeds 10^{-5}, or over the full trajectory when no active samples exist. Cost uses the total applied control, including compensation. All values describe one deterministic trajectory per setting.

Settling time is absolute time after the designated last event, using a state-norm threshold of 0.05. The event is 17 s for actuator cases and 16 s for the sensor-only case. Scenario 3's value of 16 s therefore does not indicate delayed regulation.

5. Results across the four scenarios

The main stored experiment uses actuator amplitude 1.0 and ripple 0.10. States recover toward the origin after the actuator disturbance. Sensor-only faults leave the state and control trajectories unchanged under the exact projection. Scenario pairs 1/3 and 2/4 have identical stored state trajectories.

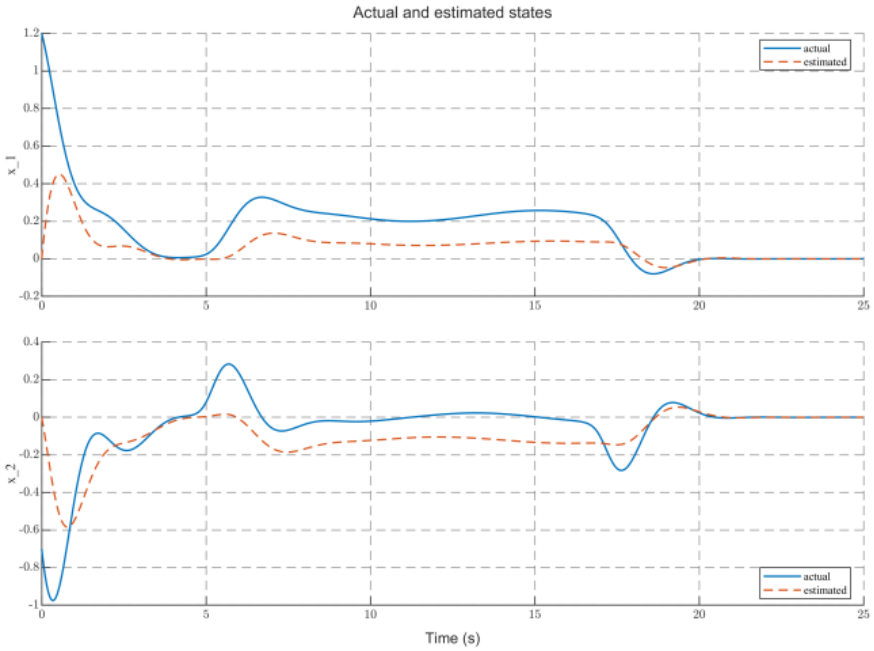


Figure 3. Actual and estimated states under simultaneous actuator and sensor faults. Reproduced from the Persian report, Figure 4-15. Sensor-fault reconstruction is shown in Figure 2 of this summary.

Table 7. Regulation and control performance, proposed mode.

Scenario	State ISE	Control energy	Cost J_T	Settling (s)
1	1.4986	1.8631	4.7114	3.60
2	2.3434	6.4087	9.9271	19.72
3	1.4986	1.8631	4.7114	16.00
4	2.3434	6.4087	9.9271	19.72

Table 8. Estimation RMSE under the metric conventions in Section 4.

Scenario	State estimate	Actuator fault	Sensor fault
1	0.1541	0.2092	0.0760
2	0.2112	0.3225	0.2139
3	0.1541	0.2092	1.970×10^{-5}
4	0.2112	0.3225	0.1977

All four archived runs keep both x and $\hat{x}$ within the design square at the recorded samples. The peak state norm is 1.4015 and the peak control magnitude is 1.4594 in each case. In scenario 3, the maximum active-interval sensor reconstruction error is 8.60×10^{-5}. In scenario 4, its active RMSE rises to 0.1977 because the actuator fault creates state-estimation error. Nonzero fault-free RMSE values mainly reflect the initial observer transient.

6. Comparison with uncompensated and ideal modes

The uncompensated mode uses the same learned policy and adaptive observer, with the actuator compensation term disabled. The ideal-information mode evaluates the learned policy at the true state and uses the exact actuator fault for compensation. It provides a reference with privileged information; its nominal policy remains the learned policy.

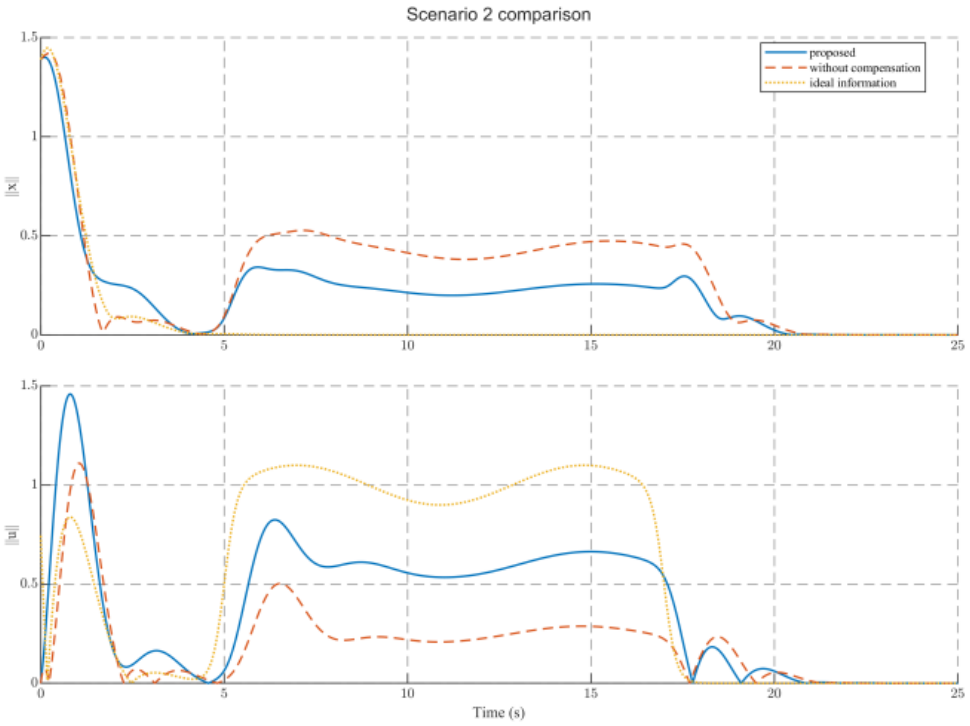


Figure 4. Actuator-fault baseline comparison at amplitude 1.0 and ripple 0.10. Reproduced from the Persian report, Figure 4-10. The blue, dashed orange, and dotted yellow curves denote proposed, uncompensated, and ideal-information modes.

Table 9. Matched baseline comparison, scenario 2.

Mode	State ISE	Control energy	Cost J_T	Settling (s)
Proposed	2.3434	6.4087	9.9271	19.72
Without compensation	4.2635	2.0182	13.9360	19.99
Ideal information	1.6986	12.8868	9.9288	17.00

Values recalculated from the saved trajectories agree with the report, Table 4-5, to the displayed precision. Settling values are absolute times under the implemented event convention.

Interpretation

Relative to the uncompensated case, the proposed mode reduces state ISE from 4.2635 to 2.3434 (approximately 45.0%) and cost from 13.9360 to 9.9271 (approximately 28.8%). Control energy rises from 2.0182 to 6.4087. The benefit therefore reflects a tradeoff between regulation and actuation effort, rather than a simultaneous reduction of both.

The ideal-information reference gives the smallest state ISE, but its control energy is 12.8868 and its total cost is approximately the same as the proposed result. Exact fault cancellation does not by itself minimize the finite-horizon objective defined here. The compensator also reacts transiently to initial estimation error, even before an actual fault is present.

The report rounds its headline reductions to 45.04% and 28.77%. Recalculation from the unrounded archive gives 45.0348% and 28.7663%; the small difference in the first displayed percentage reflects rounding.

7. Sensitivity to actuator-fault amplitude

The separate saved sweep compares proposed and uncompensated control at five amplitudes. The observer and learned policy are reused across the amplitude settings. In the current sweep implementation, ripple amplitude equals (0.10/0.70) times nominal amplitude; therefore, at amplitude 1.0 the ripple is 0.142857, distinct from the principal run's 0.10.

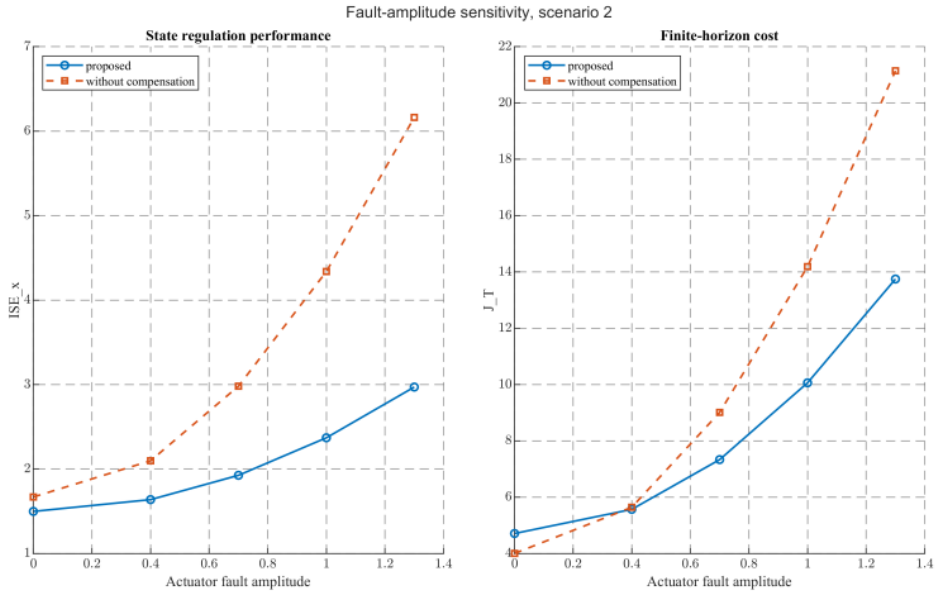


Figure 5. Original scenario-2 amplitude-sensitivity figure, reproduced from the Persian report, Figure 4-19. Its plotted trends agree with the saved sweep; the report's numerical sweep table differs as noted below.

Table 10. Saved sweep results; P = proposed, NC = no compensation.

Amplitude	ISE P	ISE NC	ISE reduction	J_T P	J_T NC	J_T reduction
0.0	1.4986	1.6692	10.22%	4.7114	4.0043	-17.66%
0.4	1.6380	2.0982	21.93%	5.5685	5.6429	1.32%
0.7	1.9259	2.9809	35.39%	7.3354	9.0111	18.60%
1.0	2.3703	4.3385	45.37%	10.0633	14.1891	29.08%
1.3	2.9707	6.1625	51.79%	13.7485	21.1438	34.98%

At amplitude 1.3, state ISE is reduced by 51.79% and cost by 34.98% in this saved sweep. At zero amplitude, ISE improves by 10.22% but cost increases by 17.66%, owing to control effort associated with observer transients. All saved sweep rows report in-domain trajectories and no saturation; the configured input limit is infinite.

Table 11. Numerical provenance: three distinct configurations.

Configuration	Amplitude / ripple	Proposed ISE	Proposed J_T
Principal archive: results/1/	1.0 / 0.10	2.3434	9.9271
Top-level results archive	1.3 / 0.10	2.9032	13.4041
Saved sweep at amplitude 1.0	1.0 / 0.142857	2.3703	10.0633

The report's sweep table gives slightly different values (at amplitude 1.3: ISE 2.9255, cost 13.5189, reductions 51.54% and 34.76%). Those entries do not match the supplied sweep CSV. This summary uses the traceable saved CSV for Figure 5 and Table 10, while preserving the distinction from the report.

8. Stability scope, reproducibility, and conclusions

8.1 Conditional stability result

The report constructs a composite Lyapunov function for plant regulation and observer errors. With $w = [x^T, e_x^T, \tilde{f}_a^T]^T$, the analysis has the form

$$\mathcal{V} = V^*(x) + \ell_o(e_x^T P e_x + \tilde{f}_a^T \Gamma^{-1} \tilde{f}_a)$$

$$\dot{\mathcal{V}} \le -\kappa_c \mathcal{V} + \chi_c, \qquad \limsup_{t\to\infty} \|w(t)\| \le r_c$$

Uniform ultimate boundedness requires the structural assumptions, a feasible observer design, an admissible final policy, a separate uniform policy-error bound, unsaturated control, and a suitable invariant sublevel set within the domain where both x and $\hat{x}$ remain valid. The executable policy test covers a finite grid. It does not compute the policy-error bound or certify the continuous-domain invariant set. The theoretical radius p.theory.rc is NaN by default; aggregate error plots are numerical diagnostics.

8.2 Implementation and reproduction

The report specifies MATLAB R2024a. YALMIP and an SDP solver such as SDPT3 are external dependencies for the solver-based observer route and are not bundled with the 13 source files. The analytical fallback supports the supplied example. No external dataset, pretrained weights, or Simulink model is required. Run *main_chapter4* for the main study and *plot_fault_amplitude_sweep* separately for sensitivity analysis.

The supplied implementation uses a known model, exact measurements, two-state polynomial features, and a two-dimensional grid. The noiseStd parameter is declared but does not inject noise. Finite input clipping is available, while the documented theorem assumes unsaturated control. Broader applications require redesign of the model, projection, critic/grid, and fault channels. Fresh MATLAB execution could not be completed because the application failed during startup; this document analyzes existing saved results.

8.3 Conclusions

The matched archive supports the project's principal finding: adaptive actuator-fault compensation substantially improves state regulation and weighted cost in the tested faulty case, at higher control energy. The virtual-output construction isolates the modeled sensor fault from the feedback dynamics, and sensor-only reconstruction is accurate after the observer transient. During overlapping faults, sensor reconstruction becomes less accurate because it depends on state estimation. The evidence supports this specific deterministic demonstration and its assumptions; it does not establish global optimality or performance for general faulty nonlinear plants.